BUSINESS PROPOSAL

UMHackathon 2026

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1. Executive Summary:

Motor insurance claims in Malaysia remain highly manual, document-heavy, and time-consuming. Claims officers are required to review multiple documents such as police reports, policy cover notes, identification documents, vehicle photos, and workshop quotations before making a decision. This process is not only slow but also inconsistent, leading to delays, operational inefficiencies, and potential claim leakage.

SettleOps AI is designed to transform this process into a structured, intelligent workflow. Instead of manually piecing together information from multiple sources, claims officers are supported by an AI-powered decision cockpit that automates document intake, extracts key evidence, and analyzes claims through multiple specialized reasoning layers. The system consolidates all findings into a transparent and structured view, allowing officers to understand, validate, and act on decisions more efficiently.

At the core of SettleOps AI is a multi-agent architecture that mirrors how real insurance decisions are made. Different AI agents independently evaluate policy coverage, liability, damage, fraud risk, and payout logic, before consolidating their findings into a unified recommendation. This approach ensures that decisions are not only faster but also more consistent and explainable. Every step of the decision process is traceable, enabling insurers to maintain auditability and compliance while benefiting from automation.

Rather than replacing human judgment, SettleOps AI enhances it. Claims officers remain in control through a human-in-the-loop system, where they can approve, reject, or challenge specific parts of the AI's analysis. This creates a collaborative environment where AI accelerates decision-making while preserving professional oversight and accountability.

The platform is positioned as a decision-support solution for insurers, takaful operators, claims administrators, and adjusters who handle high volumes of motor claims. As claim complexity increases and operational pressure grows, there is a clear need for solutions that improve efficiency without compromising accuracy or control. SettleOps AI addresses this gap by providing a unified system that connects document processing, intelligent analysis, and decision execution in one seamless workflow.

In essence, SettleOps AI is not just a document processing tool, but a complete claims decision platform. It enables insurers to move from fragmented, manual processes to a streamlined, intelligent system that delivers faster turnaround, better consistency, and greater transparency in claim decisions.

2. Product Overview / Solution

Key Features:

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SettleOps AI is an AI-powered claims decision cockpit. It is designed to support the full motor accident claim journey from document submission to officer approval and repair approval letter generation.

Feature	Description	Business Benefit
Claim Case Intake	Creates claim cases and allows users to upload all relevant documents such as police reports, policy documents, identification records, vehicle photos, and workshop quotations.	Centralizes all claim evidence and reduces fragmented document handling.
AI Document Extraction	Automatically extracts key information from PDFs, images, and other documents.	Eliminates manual data entry and accelerates initial claim review.
Automatic Evidence Tagging	Classifies uploaded documents into categories such as police report, policy cover note, workshop quotation, and identification documents.	Detects missing or incomplete evidence early, reducing delays.
Multi-Agent Claim Analysis	Analyzes claims across multiple aspects including policy coverage, liability, damage, fraud risk, and payout.	Produces structured, consistent, and explainable claim decisions.
Shared Decision Blackboard	Consolidates all analysis results into structured sections for easy review.	Provides transparency and allows officers to understand how decisions are made.
Deterministic Payout Calculation	Calculates claim payouts using rule-based logic based on verified data.	Ensures accurate, auditable, and consistent financial decisions.
Human-in-the-Loop Review	Allows officers to approve, reject, or challenge AI-generated recommendations.	Maintains human control while improving efficiency.
Targeted Rerun Mechanism	Enables officers to re-evaluate specific aspects of a claim without restarting the entire process.	Saves time and improves decision precision.
Real-Time Workflow	Provides live updates on claim	Improves visibility and reduces

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Updates	processing status and analysis results.	uncertainty during decision-making.
Decision Artifact Generation	Generates structured reports and approval-ready documents.	Speeds up claim approval and supports compliance requirements.
Repair Approval PDF	Produces standardized repair approval letters with claim and cost details.	Accelerates communication with workshops and improves customer service.
Digital Signing	Enables digital signature and stamping of generated documents.	Reduces manual approval steps and supports paperless workflows.
AI Assistant (Chat)	Allows users to ask questions about claim decisions and system outputs.	Improves explainability and user confidence in AI decisions.

Technology Used:

Technology Area	Current Implementation
Backend API	FastAPI-based backend providing APIs for claim management, document handling, workflow control, and user interactions.
Workflow Orchestration	Agent-based workflow system that manages claim processing steps and supports human-in-the-loop interaction.
AI Processing Layer	Uses AI models for document understanding, image analysis, and reasoning across claim data.
Multi-Model Support	Support Gemini, OpenAI-compatible providers (e.g. Ilmu), and local LLMs (Ollama/LM Studio), with automatic failover rotation for resilience.
Document Processing	Handles PDFs, images, and office documents, extracting and structuring relevant claim data.
Real-Time Updates	Streaming mechanism that delivers live workflow updates and system outputs to users.
Storage System	Stores claim data, documents, and generated outputs with structured organization.

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PDF Generation	Automatically generates claim reports and repair approval documents.
Digital Signature	Applies digital signatures and stamps to finalized documents.
Web Integration	Web-based interface for claims officers with responsive and interactive design.
Security & Authentication	Implements user authentication, access control, and secure data handling.
Deployment	Self-hosted, Docker-based deployment packaged for on-premise installation within the insurer's own infrastructure.

3. Market Research & Market Dynamics

3.1 Problem Landscape

Motor insurance claims are highly document-intensive, time-sensitive, and operationally costly. A typical motor accident claim requires multiple documents such as police reports, vehicle photos, driver identification, policy cover notes, road tax or registration details, workshop quotations, and adjuster reports. These documents often come in different formats and quality levels, requiring claims officers to manually extract key information, cross-check evidence, verify policy terms, assess damage, and detect potential fraud.

This manual process creates several operational challenges. It slows down claim turnaround time, increases workload for claims officers, and introduces inconsistencies in decision-making. As a result, insurers face higher risks of claim leakage, fraud, and customer dissatisfaction.

The pressure on insurers continues to grow. According to industry reports, Malaysia's motor insurance segment remains the largest contributor to general insurance premiums while also experiencing increasing claim costs. Rising repair costs, higher accident frequency, and more complex vehicle technologies have contributed to underwriting pressure, with the motor segment continuing to face profitability challenges.

As claim volumes increase and operational complexity grows, insurers require solutions that can improve efficiency, standardize decisions, and reduce manual workload without compromising accuracy or compliance.

3.2 Market Niche

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SettleOps AI focuses on the motor insurance claims automation niche, starting in Malaysia before expanding to Southeast Asia. The platform targets organizations that handle high volumes of claims and require faster, more consistent decision-making processes.

Primary customer segments include general insurers and takaful operators with large motor portfolios, claims departments seeking operational efficiency, third-party administrators and loss adjusters handling outsourced claims processing, and panel workshop networks involved in repair approvals.

The initial focus is on motor own-damage and accident repair claims, particularly those within a defined complexity or value range where AI-assisted processing can deliver the greatest efficiency gains. These claims represent a significant portion of total claim volume and are often repetitive and document-heavy, making them suitable for automation.

Over time, the solution can be extended to other general insurance lines such as property, travel, personal accident, and SME claims. Regional expansion opportunities exist in markets such as Singapore,

Indonesia, Thailand, and the Philippines, where similar claims workflows and operational challenges are present.

3.3 Market Sizing

The market sizing reflects both the broader insurance landscape and the realistic revenue opportunity for claims automation solutions like SettleOps AI.

Market Tier	Definition	Estimated Value
TAM (Total Addressable Market)	Total motor insurance premium pool in Malaysia, representing the overall value of claims activity and operational processes that can be enhanced through automation.	~RM10 billion+
SAM (Serviceable Available Market)	Portion of the market consisting of insurers, takaful operators, and claims service providers that can realistically adopt digital claims processing solutions.	~RM500 million – RM1 billion
SOM (Serviceable Obtainable Market)	Realistic revenue capture for SettleOps AI within the first 18 months through pilot projects and early commercial deployments.	RM3 – RM5 million

This sizing reflects that SettleOps AI does not capture insurance premiums directly, but instead provides software and workflow solutions that improve claims processing efficiency and decision quality.

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3.4 Market Dynamics & Placement:

Rising Claims Pressure: Motor claim costs are increasing due to higher accident frequency, rising repair costs, and the growing complexity of modern vehicles. Insurers are under pressure to improve efficiency, reduce claim leakage, and accelerate claim turnaround time.

Motor Insurance Dominance: Motor insurance remains the largest segment in Malaysia's general insurance industry. The high claim volume and operational intensity make it the most suitable entry point for automation solutions.

Digital Insurance Momentum: The regulatory push for digital insurers and takaful operators in Malaysia is accelerating the adoption of technology-driven solutions. Insurers are increasingly investing in digital transformation to remain competitive.

AI Adoption in Claims Processin: Globally, insurers are adopting AI to improve claims efficiency, reduce processing time, and enhance decision accuracy. Successful implementations have demonstrated significant improvements in operational performance and customer satisfaction.

Ecosystem Readiness: Malaysia already has established digital claims infrastructure, such as existing claims management platforms and panel workshop networks. This indicates that the ecosystem is ready for further innovation, particularly solutions that enhance intelligence and decision-making rather than replacing existing systems entirely.

4. Value Proposition & Competitive Advantage

4.1 Value Proposition

SettleOps AI delivers value by transforming manual, fragmented claim review into a structured and intelligent decision workflow. The platform enhances efficiency, consistency, and transparency across the claims process while maintaining full human control.

For Insurers
SettleOps AI reduces the time required to prepare claim decisions by automating document processing, policy validation, and payout calculations. This improves operational efficiency, lowers manual workload, and reduces claim leakage caused by inconsistent or rushed decisions. The platform also strengthens fraud detection by identifying inconsistencies across multiple documents and ensures compliance through complete audit trails and decision transparency.

For Claims Officers
The system provides a centralized decision cockpit where officers can clearly view all relevant claim

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information, including policy analysis, liability assessment, damage evaluation, fraud indicators, and payout recommendations. Officers can interact with the system, understand how decisions are formed, and challenge specific components without restarting the entire process. This improves productivity, reduces cognitive load, and increases confidence in decision-making.

For Customers and Workshops

SettleOps AI enables faster claim processing and quicker repair approvals. By reducing delays and improving decision clarity, customers experience a smoother claims journey, while workshops benefit from faster authorization and reduced administrative friction.

4.2 Competitive Advantage

Factor	Competitor	SettleOps
End-to-End Claim Workflow	ABBYY – Focuses on document extraction and classification only	Full workflow from document intake → analysis → decision → report → signing
Final Claim Report Generation	Kofax – No built-in approval document generation	Automatically generates approval-ready claim reports and repair letters
Digital Signature Integration	OpenText – Requires external tools for signing	Built-in digital signature and stamping capability
AI-Based Decision Making (Policy, Liability, Fraud)	UiPath – Focuses on automation, lacks deep decision intelligence	Multi-agent AI evaluates policy, liability, damage, fraud, and payout
Multi-Agent Architecture	Rossum – Single pipeline AI processing	Multiple specialized agents working collaboratively

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Deterministic Payout Calculation	Infrrd – No built-in payout logic	Rule-based payout calculation for accuracy and auditability
AI Auditor / Consistency Checking	Hyperscience – No cross-validation layer	Auditor agent ensures consistency across all decision components
Human-in-the-Loop with Smart Rerun	Kofax – Requires manual reprocessing	Targeted rerun of specific decision components
Explainability (Transparent Decision Process)	Amazon Web Services (Textract) – Limited explainability	Structured decision blackboard showing full reasoning
Document Classification (Insurance-Specific)	Google Cloud (Document AI) – Generic classification	Claim-specific evidence tagging and validation
Real-Time Workflow Updates	ABBYY – Batch processing	Live workflow updates and decision tracking
Chat-Based Interaction (Case-Aware)	OpenText – No conversational interface	Case-aware AI assistant for decision explanation
Fraud Detection Integration	Rossum – Not specialized for fraud detection	Dedicated fraud analysis component
Insurance-Specific Workflow Design	Hyperscience – Generic document AI	Designed specifically for insurance claims

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One-Click Usability (Non-technical users)	Amazon Web Services – Requires developers	Ready-to-use system

5. Competitor Analysis (Porter's Five Forces)

Force	Intensity	Assessment
Competitive Rivalry	High	The insurance software market includes established claims platforms such as Merimen, Guidewire, Duck Creek Technologies, and AI damage assessment tools. SettleOps AI differentiates through agentic reasoning, decision blackboard, and local motor claim focus.
Threat of New Entrants	Medium	AI prototypes are easier to build today, but production insurance deployment requires domain expertise, integrations, auditability, data security, and trust.
Bargaining Power of Buyers	High	Insurers are sophisticated enterprise buyers with long procurement cycles, strong compliance requirements, and multiple vendor options.
Bargaining Power of Suppliers	Medium	SettleOps depends on LLM API providers (Gemini, Ilmu), OCR libraries and optional local model infrastructure. Risk is reduced through multi-provider failover and local LLM support.
Threat of Substitutes	Medium	Manual claims teams, outsourced adjusters, rule-based workflows, and existing claim systems can substitute some functionality.

Competitor Comparison Table:

Competitor	Strength	Weakness	SettleOps AI Advantage
ABBYY	Industry-leading OCR and document extraction accuracy	No end-to-end claims workflow; lacks report generation and signing	Complete pipeline from document upload to decision, report generation,

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		capabilities	and digital signing
Kofax	Strong workflow automation and enterprise integration	Complex implementation; lacks built-in claim decision logic and final output generation	Integrated multi-agent decision system with automated approval document generation
OpenText	Robust document management and compliance features	Focused on storage and management rather than intelligent decision-making	AI-driven claim analysis with transparent audit trail and explainable outputs
UiPath	Highly flexible automation using RPA and AI	Requires technical expertise; lacks insurance-specific decision intelligence	Ready-to-use, domain-specific system designed for insurance claim workflows
Amazon Web Services (Textract)	Powerful document data extraction capabilities	Only provides extraction; no workflow, decision logic, or output generation	End-to-end system including reasoning, payout calculation, and report generation
Google Cloud (Document AI)	Advanced AI-based document understanding	Generic solution without insurance-specific workflow or outputs	Insurance-focused document classification and structured decision pipeline
Infrrd	Specialized in insurance document extraction	Does not provide payout calculation or approval document generation	Deterministic payout engine and approval-ready document output
Hyperscience	High accuracy for complex and handwritten documents	No workflow automation or claim decision capabilities	Integrated liability, fraud, policy analysis, and audit validation
Rossum	Modern cloud-based document processing	Limited to extraction and validation; not claim-specific	Multi-agent architecture with explainable decision system and audit trail
Quantic Solutions	Local customization and workflow solutions	Not specialized in insurance claims; requires custom development	Fully built, insurance-specific claim automation platform ready for deployment

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6. Business Model Canvas

6.1. Customer Segmentation

SettleOps AI targets organizations involved in high-volume motor insurance claims processing, where efficiency, accuracy, and turnaround time are critical.

1. **General Insurers**
   Insurers with large motor portfolios that face high claim volumes and operational pressure to reduce claim cycle time, improve consistency, and control claim leakage.
2. **Takaful Operators**
   Operators require Shariah-compliant insurance processes while facing similar challenges in claim processing efficiency and decision consistency.
3. **Digital Insurers**
   New and emerging digital-first insurers in Malaysia under Bank Negara Malaysia's digital insurance framework, seeking scalable and automated claim processing solutions.
4. **Third-Party Administrators (TPAs)**
   Organizations that manage claims on behalf of insurers and require tools to improve productivity, standardize decisions, and reduce manual workload.

6.2 Revenue Stream

Primary: Perpetual or Annual Software License Fee (per site/entity)

One-time (perpetual) or recurring annual licensing fee charged to insurers based on deployment scope (e.g., per branch, region, or organization). This forms the core revenue stream without reliance on claim volume tracking.

Existing & Retained Revenue Streams:

- **Implementation & Integration Fee:** One-time fee covering system deployment, infrastructure setup, workflow configuration, and integration with existing insurer systems.
- **Annual Maintenance & Support Contract:** Recurring annual fee for software updates, bug fixes, technical support, and service-level agreement (SLA) coverage.

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- **Premium Analytics Module:** Optional add-on license providing advanced analytics such as fraud detection, claim leakage insights, and performance monitoring.
- **Digital Signing & Artifact Add-On:** Optional module for generating digitally signed approval documents, maintaining audit trails, and supporting compliance reporting.
- **Professional Services:** Project-based services including customization, insurer-specific rule configuration, regulatory compliance support, and staff training.

6.3. Cost Structure

Cost Drivers	Cost Type
Engineering & Product Development	Fixed Cost
Software Maintenance & Update	Fixed Cost
On-Site Implementation Labour	Variable Cost
AI Model & OCR Processing	Variable Cost
Security & Compliance	Fixed & Overhead
Enterprise Sales & Onboarding	Overhead
Customer Support	Overhead & Variable

6.4. Key Partners

Partner Type	Potential Partners
Insurer Partners	General insurers and takaful operators in Malaysia
Claims Ecosystem Partners	Loss adjusters, third-party administrators, and panel workshop networks

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Existing Claims Platforms	Merimen / Fermion-style systems where SettleOps AI acts as an intelligence layer
AI Providers	Google Cloud (Gemini), Ilmu, local LLM providers, OCR vendors
Regulatory & Compliance Advisors	Insurance compliance consultants, legal advisors, and cybersecurity auditors
Payment & Finance Ecosystem	Future integration with insurer payment systems for automated payout workflows, subject to compliance

6.5 Risk Assessment

Risk Type	Impact	Likelihood	Mitigation
AI hallucination or incorrect reasoning	High	Medium	Maintain human approval, apply deterministic payout logic, provide transparent evidence and audit trails, and implement confidence thresholds
Data privacy and cybersecurity	High	Medium	Implement encryption, role-based access control, audit logs, secure storage, and align with regulatory requirements
Enterprise integration delays	Medium	High	Start with standalone deployment, then integrate progressively through APIs
Regulatory concerns	High	Medium	Position as decision-support system with mandatory human approval and full auditability
LLM vendor dependency	Medium	Medium	Use multi-provider architecture, caching, and local model fallback
Low-quality or missing documents	Medium	High	Implement document validation, evidence tagging, and missing-document detection workflows

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Claims officer adoption resistance	Medium	Medium	Design as augmentation tool, provide explainability, and conduct training with pilot feedback
Competitor response	Medium	Medium	Focus on specialization in motor claims, fast deployment, and strategic partnerships

7. Implementation & Financials

7.1 Go-to-Market Strategy

SettleOps AI adopts a **B2B enterprise-first strategy**, focusing on pilot-based adoption rather than a direct large-scale rollout. This approach reduces risk for insurers while allowing the platform to demonstrate measurable value before full deployment.

The initial market focus is on **Malaysian motor insurance**, where claim workflows are standardized, document-heavy, and high in volume. This creates an ideal environment to validate efficiency improvements and decision consistency.

The go-to-market sequence is structured as follows:

1. **Pilot Phase:** Collaborate with one insurer, takaful operator, third-party administrator, or loss adjuster using anonymized historical claims. The objective is to validate document extraction accuracy, decision consistency, payout calculation logic, and overall workflow performance.
2. **Performance Validation:** Measure key metrics such as reduction in claim preparation time, detection of missing documents, consistency of payout calculations, and user satisfaction among claims officers.
3. **Controlled Live Deployment:** Expand to selected live claims under full human supervision. This ensures operational safety while demonstrating real-world impact.
4. **Commercial Conversion:** Convert pilot partners into paying enterprise customers by demonstrating measurable efficiency gains and operational value.
5. **Expansion & Integration:** Integrate with existing claims systems and expand to additional insurers, adjusters, and related claim ecosystems.

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7.2 Implementation Plan

Phase	Action	Timeline	Target
Phase 1: Pilot Validation	Deploy sandbox with anonymized historical claims, validate document extraction, agent outputs, payout logic, and officer cockpit	Month 1–3	1 pilot partner, 500–2,000 historical claims tested
Phase 2: Controlled Live Pilot	Process selected live claims under human review; integrate document upload, decision review, PDF output, and officer challenge flow	Month 4–6	3,000–5,000 live claims processed, ≥40% reduction in review preparation time
Phase 3: Commercial Launch	Convert pilot into paid deployment; add enterprise features such as admin controls, security enhancements, analytics, and integration APIs	Month 7–12	3 paying enterprise customers, 20,000 claims processed per month
Phase 4: Scale Malaysia	Expand to additional insurers, TPAs, and adjusters; introduce compliance dashboards and advanced fraud analytics	Month 13–18	6–8 enterprise customers, 60,000–80,000 claims processed per month
Phase 5: Regional Expansion	Adapt workflows and templates for other Southeast Asian markets	Month 19–30	First regional pilot and validated expansion model

7.3 Revenue Projections (3-Year Estimate)

Assumptions:

Revenue growth is driven by increasing enterprise adoption, higher claim volumes, and expanded platform usage. Subscription revenue scales with the number of enterprise customers, while per-claim revenue increases with processing volume. Setup and integration revenue is higher in early stages and gradually becomes a smaller proportion as recurring revenue grows.

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Metric	Year 1	Year 2	Year 3
Enterprise Customers	3	8	15
Claims Processed Annually	120,000	600,000	1,800,000
Base Annual License Revenue	RM1.2M	RM3.5M	RM7.5M
Volume-Tier Uplift Revenue	RM0.3M	RM1.5M	RM4.5M
Setup / Integration / Services Revenue	RM0.6M	RM1.4M	RM2.0M
Premium Module Revenue	RM0.3M	RM1.0M	RM2.5M
Total Revenue	**RM2.7M**	**RM9.9M**	**RM25.5M**
Estimated Operating Cost	RM4.2M	RM7.5M	RM16.0M
Net Position	**RM(1.5M)**	**RM2.4M**	**RM9.5M**

7.4 Key Financial Milestones

- **Month 4:** Complete first pilot using anonymized historical claims
- **Month 6:** Complete controlled live pilot with measurable efficiency improvement
- **Month 9:** Secure first paying enterprise customer
- **Month 12:** Reach 3 enterprise customers and approximately RM2.7M annualized revenue
- **Month 18:** Expand to 6–8 enterprise customers and scale to 60,000–80,000 claims per month
- **Month 24:** Achieve positive operating contribution driven by recurring revenue
- **Month 30:** Launch first regional pilot in Southeast Asia

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